

Design Method of Campus Fire Outdoor Evacuation Plan: Taking Beijing 101 Middle School as an Example



北京一零一中
BEIJING 101 HIGH SCHOOL

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Under the influence of fire hazard gas release and wind field, how to choose safer outdoor evacuation routes and assembly points on campus?

Abstract: Campus fire evacuation often overlooks outdoor toxic smoke movement. This study develops a GIS-based framework for safer campus evacuation by integrating hazardous-gas release, wind-driven advection-diffusion, and route optimization. Using Beijing 101 Middle School, the model estimates **CO**, **HCN**, and **HCl** emissions, simulates time-varying gas concentration fields, and selects routes based on toxic exposure, distance, road width, and assembly-point capacity. Results show safer routes may differ from shortest paths under changing plume and capacity conditions.

Keywords: Campus fire evacuation; Hazardous gas diffusion; Wind-driven advection-diffusion; FED exposure; GIS road network; Route optimization.

1 Overall Model Framework

The overall framework of this study is a connected models chain: begin at fire chemistry, going to hazardous gas diffusion, and finally evacuation scheme optimization.

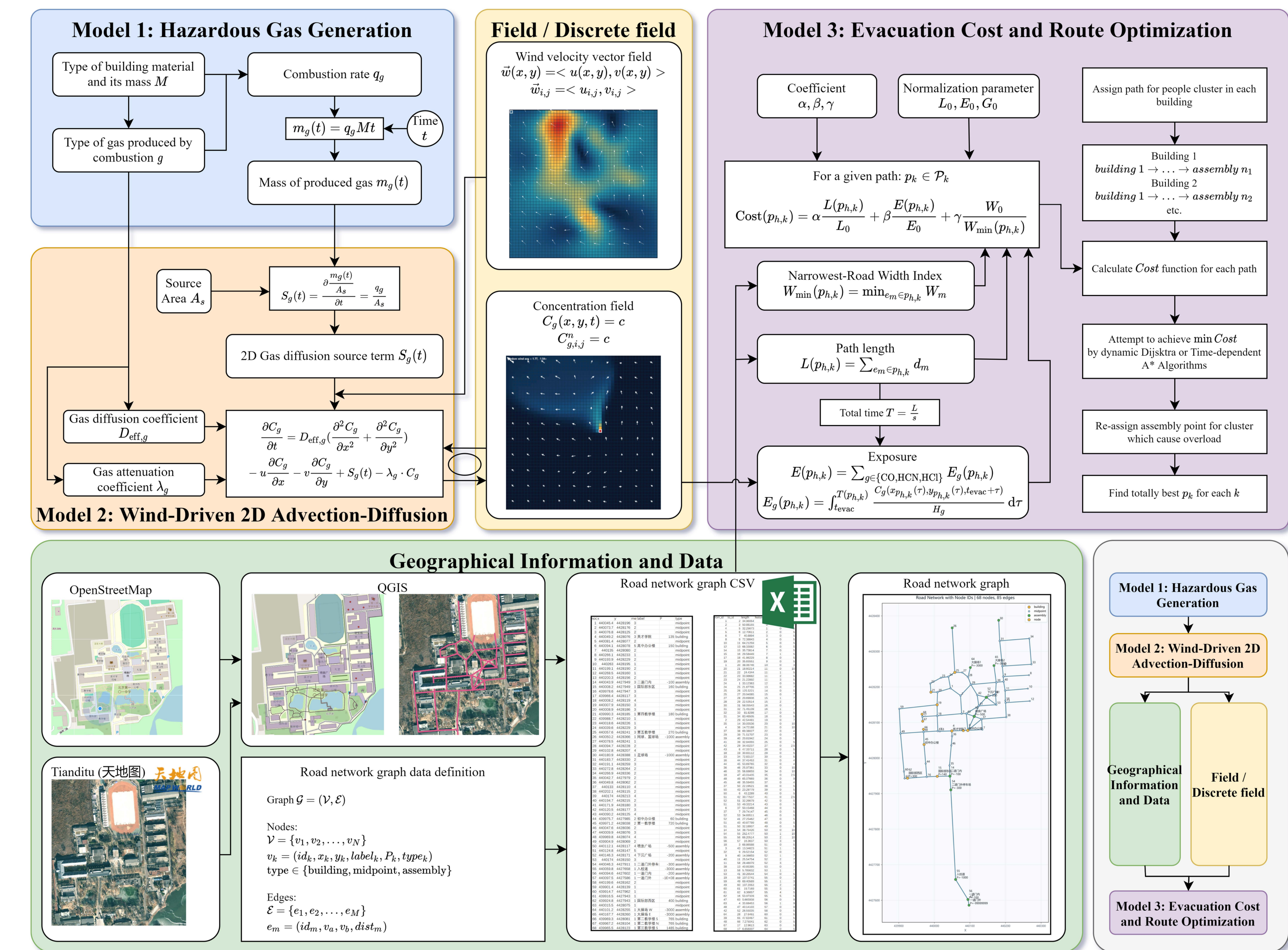


Figure 1. Model Schema Framework

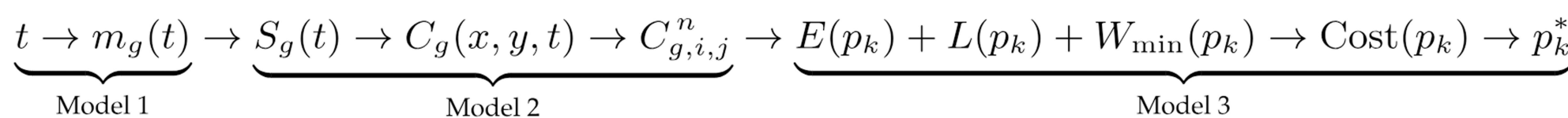


Figure 2. Model Variable Chain

2 Data Sources and Preprocessing

2.1 Geographical Map

The geographical map of the school can be obtained from geographical data supplier, such as OpenStreetMap, which provides detailed information about the campus layout. The map data is processed to extract the necessary features for modeling. Geographical features are abstracted into a graph structure, denoted as $\mathcal{G}$ and defined as:

The overall structure and data of the required graph $\mathcal{G}$ is shown below:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E})$$

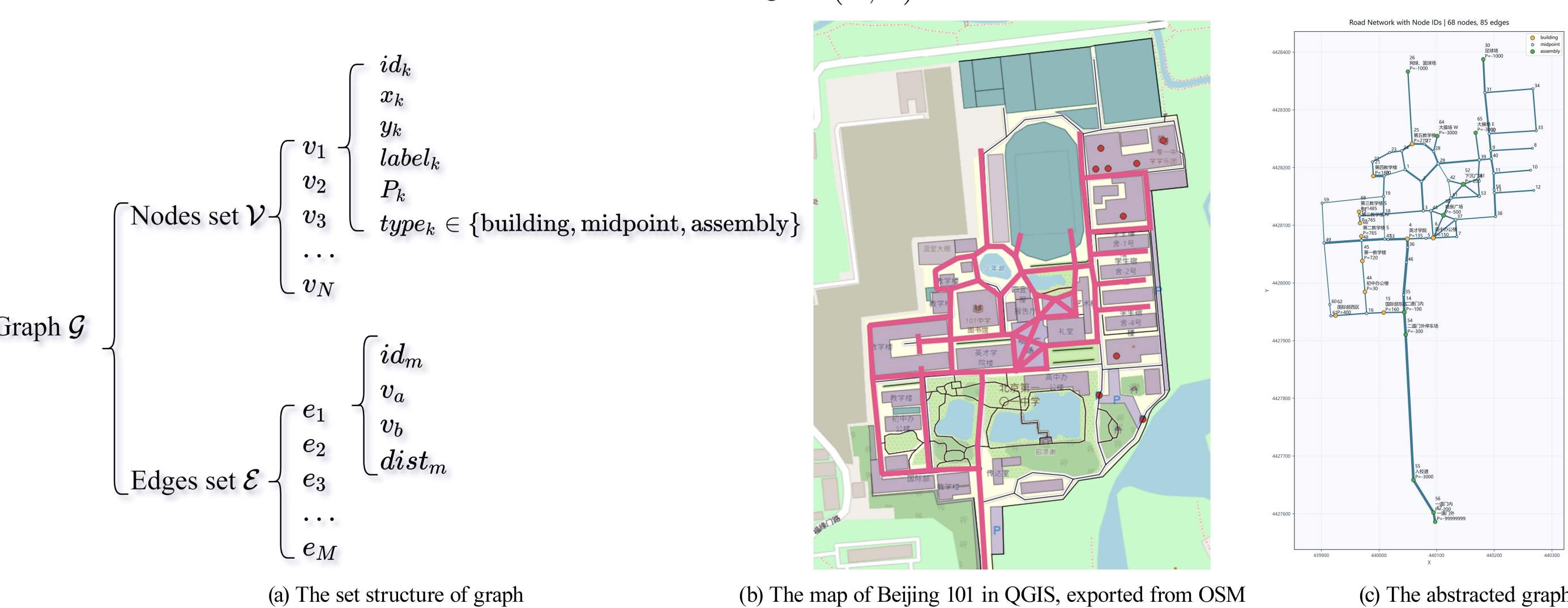


Figure 2. Geographical data processing

(a) Complete processed GIS node table						(b) Complete processed GIS node table (Continued)						(c) Complete processed GIS edge table					
node_id	x ¹	y ¹	merged	label	P	type ²	node_id	x ¹	y ¹	merged	label	P	type ²	node_id	x ¹	y ¹	merged
1	440045.433	4428106.377	2	—	midpoint	—	37	440132.958	4428109.72	4	—	midpoint	—	38	440202.134	4428115.04	2
2	440073.879	4428115.736	2	—	midpoint	—	38	440202.134	4428115.04	2	—	midpoint	—	39	440174.424	4428123.03	4
3	440076.821	4428124.991	2	—	midpoint	—	39	440174.424	4428123.03	4	—	midpoint	—	40	440194.703	4428123.23	2
4	440049.183	4428076.323	3	英才学院	135	building	40	440194.703	4428123.23	2	—	midpoint	—	41	440173.917	4428180.338	3
5	440081.822	4428077.41	2	—	midpoint	—	41	440173.917	4428180.338	3	—	midpoint	—	42	440210.548	4428177.36	3
6	440094.115	4428077.991	5	—	midpoint	—	42	440210.548	4428177.36	3	—	midpoint	—	43	440090.166	4428125.163	4
7	440134.952	4428080.045	2	高中办公楼	150	assembly	43	440090.166	4428125.163	4	—	midpoint	—	44	439975.689	4427984.603	2
8	440056.149	4428233.373	1	—	midpoint	—	44	439975.689	4427984.603	2	初中办公楼	60	building	45	439971.172	4428038.11	2
9	440039.877	4428239.277	2	—	midpoint	—	45	439971.172	4428038.11	2	—	midpoint	—	46	440047.566	4428035.724	2
10	440263.942	4428195.421	1	—	midpoint	—	46	440047.566	4428035.724	2	—	midpoint	—	47	440009.891	4428076.092	3
11	440159.053	4428190.061	2	—	midpoint	—	47	440009.891	4428076.092	3	—	midpoint	—	48	439969.823	4428073.679	4
12	440268.53	4428186.666	1	—	midpoint	—	48	439969.823	4428073.679	4	—	midpoint	—	49	439964.349	4427910.54	1
13	440200.345	4428155.849	2	—	midpoint	—	49	439964.349	4427910.54	1	二校门外停车场	-300	assembly	50	440046.349	4427910.54	1
14	440051.021	4427949.318	3	二校门内	-100	assembly	50	440046.349	4427910.54	1	—	midpoint	—	51	440124.799	4428146.88	5
15	440008.188	4427948.619	1	—	midpoint	—	51	440124.799	4428146.88	5	—	midpoint	—	52	440146.308	4428170.932	4
16	439978.852	4427947.304	3	—	midpoint	—	52	440146.308	4428170.932	4	—	midpoint	—	53	440174.028	4428150.067	3
17	439966.422	4428116.587	3	—	midpoint	—	53	440174.028	4428150.067	3	—	midpoint	—	54	440046.349	4427910.54	1
18	440008.21	4428107.073	4	—	midpoint	—	54	440046.349	4427910.54	1	—	midpoint	—	55	440039.803	4427638.41	1
19	440007.802	4428149.693	3	—	midpoint	—	55	440039.803	4427638.41	1	—	midpoint	—	56	440046.349	4427910.54	1
20	440008.34	4428185.543	3	—	midpoint	—	56	440046.349	4427910.54	1	—	midpoint	—	57	440097.509	4427585.991	1
21	439992.318	4428185.433	1	—	midpoint	—	57	440097.509	4427585.991	1	—	midpoint	—	58	440199.388	4428161.385	2
22	439988.67	4428200.801	1	—	midpoint	—	58	440199.388	4428161.385	2	—	midpoint	—	59	439991.434	4428138.527	1
23	440018.564	4428205.971	1	—	midpoint	—	59	439991.434	4428138.527	1	—	midpoint	—	60	439914.652	4427962.447	1
24	440039.578	4428228.979	3	—	midpoint	—	60	439914.652	4427962.447	1	—	midpoint	—	61	439916.455	4427942.813	1
25	440057.581	4428241.054	3	—	midpoint	—	61	439916.455	4427942.813	1	—	midpoint	—	62	439924.8	4427943.414	1
26	440050.2	4428366.359	1	—	midpoint	—	62	439924.8	4427943.414	1	—	midpoint	—	63	440015.512	4428075.179	1
27	440078.52	4428240.794	1	—	midpoint	—	63	440015.512	4428075.179	1	—	midpoint	—	64	440101.247	4428254.745	1
28	440094.701	4428277.885	2	—	midpoint	—	64	440101.247	4428254.745	1	—	midpoint	—	65	440167.656	4428260.243	1
29	440102.752	4428206.837	4	—	midpoint	—	65	440167.656	4428260.243	1	—	midpoint	—	66	439969.311	4428080.917	1
30	440180.888	4428187.725	1	—	midpoint	—	66	439969.311	4428080.917	1	—	midpoint	—	67	439967.236	4428103.651	1
31	440183.741	4428239.739	2	—	midpoint	—	67	439967.236	4428103.651	1	—	midpoint	—	68	439965.541	4428123.366	1
32	440191.138	4428258.672	3	—	midpoint	—	68	439965.541	4428123.366	1	—	midpoint	—				
33	440272.808	4428365.765	2	—	midpoint	—											
34	44206.942	4428316.38	2	—	midpoint	—											
35	440042.743	4427979.52	2	—	midpoint	—											

¹ Coordinate pairs are under geocentric coordinate system of EPSG:32610, which unit is meter.
² Detailed definitions are as content.

2.2 Wind Field

To simplify the model, the wind field can be assume that is uniform across the campus, which means that the wind speed and direction are the same at all locations. The wind field is denoted as:

$$\vec{w} = \langle u, v \rangle$$

4 Model 1: Hazardous Gas Generation Model

4.1 Building Material M_b and Its Combustion Product

The model 1 considers the hazardous gases generated by combustion of common building materials, which are shown below.

Material (Formula)	Typical Locations	Combustion Products
Polyvinyl (PVC) ^{1,2}	Chloride Electrical cables, vinyl flooring, pipes	CO(g), HCl(g)
Polyurethane (PU) ³	Foam Chair cushions, sofas, insulation	CO(g), HCN(g), NO _x (g)
Polystyrene (PS) ⁴	Packaging, insulation boards	Styrene(g), CO(g)
Polyethylene (PE) ⁴	Cable jackets, containers, plastic products	CO(g), Hydrocarbons(g)
Polypropylene (PP) ⁴	Chairs, storage boxes, laboratory equipment	CO(g), Hydrocarbons(g)
Recycled Board ⁴	Desks, cabinets, partitions	CO(g), HCHO(g), PM(s)
Synthetic (SBR) ⁴	Flooring, mats, sealing strips	CO(g), SO _x (g)

Table 2. Common combustible materials in school buildings and their main products

4.2 Generation Rate q_g of gas g

After simplified assumption, The generation rate of gas g and the total mass of gas $g(t)$ respect to time t can be expressed as:

$$\frac{dm_g(t)}{dt} = q_g M_g \quad m_g(t) = \int_0^t q_g M_g dt = q_g M_g t$$

5 Model 2: Wind-Driven 2D Advection-Diffusion Model

5.1 Source Term Construction

The area-averaged source strength is defined as:

$$S_g(t) = \frac{1}{A_s} \frac{dm_g(t)}{dt} = \frac{q_g M_g}{A_s}$$

5.2 Fick's Law and its Expression in Advection-Diffusion Equation

Fick's law is the foundation of the diffusion model. In this model, the diffusion process is modeled as a wind-driven advection-diffusion process. The governing equation and its discrete expression by finited differences method is:

$$\frac{\partial C_g}{\partial t} = D_{\text{eff},g} \left(\frac{\partial^2 C_g}{\partial x^2} + \frac{\partial^2 C_g}{\partial y^2} \right) - u \frac{\partial C_g}{\partial x} - v \frac{\partial C_g}{\partial y} + S_g(x, y, t) - \lambda_g \cdot C_g$$

$$C_{g,i,j}^{n+1} = C_{g,i,j}^n + \Delta t \left[D_{\text{eff},g} \left(\frac{C_{g,i+1,j}^n - 2C_{g,i,j}^n + C_{g,i-1,j}^n}{\Delta x^2} + \frac{C_{g,i,j+1}^n - 2C_{g,i,j}^n + C_{g,i,j-1}^n}{\Delta y^2} \right) - u \delta_x^{\text{up}} C_{g,i,j}^n - v \delta_y^{\text{up}} C_{g,i,j}^n + S_{g,i,j}^n - \lambda_g C_{g,i,j}^n \right]$$

The values of $D_{\text{eff},g}$ for chosen gases are shown below:

Table 4. Typical Effective Diffusion Coefficients of Key Hazardous Gases in Air

Gas	MIN.	MAX.	TYP.
Carbon Monoxide (CO)	1.8×10^{-5}	2.4×10^{-5}	2.1×10^{-5}
Hydrogen Cyanide (HCN)	1.3×10^{-5}	1.9×10^{-5}	1.6×10^{-5}
Hydrogen Chloride (HCl)	0.9×10^{-5}	1.3×10^{-5}	1.1×10^{-5}

6 Model 3: Evacuation Cost and Route Optimization Model

6.1 Candidate Path Set

For a people cluster k , the starting point is the building node where the cluster is located, and the destination is one of the feasible assembly points. The set of candidate paths for cluster k is denoted as:

$$\mathcal{P}_k = \{p_{1,k}, p_{2,k}, \dots, p_{h,k}, \dots\}$$

6.2 Path Length and Travel Time

The route length is calculated by summing the distances of all road edges contained in the path:

$$L(p_{h,k}) = \sum_{e_m \in p_{h,k}} d_m$$

Assuming that the average evacuation speed of the crowd is s , the total travel time of the candidate path is estimated by:

$$T(p_{h,k}) = \frac{L(p_{h,k})}{s}$$

6.3 FED-Based Exposure

The exposure value is calculated from the concentration field generated by Model 2. The Fractional Effective Dose (FED) method estimates toxic hazard by accumulating small exposure fractions over time. The general FED principle in simplified version and the expression of E_g , which is FED-based exposure contribution, for gas g is beside written as:

$$\text{FED}(t) = \sum_g \int_0^t \frac{C_g(\tau)}{H_g} d\tau \quad E_g(p_{h,k}) = \int_{t_{\text{evac}}}^{T(p_{h,k})} \frac{C_g(x_{p_{h,k}}(\tau), y_{p_{h,k}}(\tau), t_{\text{evac}} + \tau)}{H_g} d\tau$$

Therefore, the comprehensive FED-based exposure contribution of all chosen gas is express as:

$$E(p_{h,k}) = \sum_{g \in \{\text{CO}, \text{HCN}, \text{HCl}\}} E_g(p_{h,k})$$

6.4 Narrowest-Road Width Index

The narrowest-road width index is used to describe whether a candidate path contains a road segment with limited walkable space, defined as:

$$W_{\min}(p_{h,k}) = \min_{e_m \in p_{h,k}} W_m$$

6.5 Comprehensive Cost Function

The comprehensive cost function is used to evaluate the cost of a given path p in candidate path set for cluster k . Because $L(p_{h,k})$, $E(p_{h,k})$, and $W_{\min}(p_{h,k})$ are in different unit, they require normalized parameters to unity the weight of each factors.

$$\text{Cost}(p_{h,k}) = \alpha \frac{L(p_{h,k})}{L_0} + \beta \frac{E(p_{h,k})}{E_0} + \gamma \frac{W_0}{W_{\min}(p_{h,k})}$$

6.6 Route Optimization

For each people cluster k , the best evacuation path is the candidate path with the minimum comprehensive cost:

$$p_k^* = \arg \min_{p_{h,k} \in \mathcal{P}_k} \text{Cost}(p_{h,k})$$

7 Simulation Process and Case Study

7.1 Simulation Program



Figure 3. Simulation Program GUI

7.2 Case Study 1: Yingcai College Fire

Table 6. Evacuation Routes Plan									
Building k	Assembly Point	P _k	Capacity	L(P _k)	T(P _k)	E(P _k)	W _{min} (P _k)	Cost(P _k , k)	Status
4 英才学院	50 操场广场	135	283 / 500	58.2	75.3	0.0077	5	0.3116	ok
6 高中办公楼	50 操场广场	150	285 / 500	43.2	56.0	9.6209 × 10 ⁻¹¹	5	0.3000	ok
15 国际部东区	54 二校门外停车场	160	220 / 300	74.5	62.1	6.545 × 10 ⁻⁹	5	0.316	recovered
21 国际部教学楼	64 大操场W	180	2700 / 3000	170.6	142.2	0	5	0.3566	ok
25 第三教学楼	64 大操场W	270	2700 / 3000	69.3	57.3	0	5	0.3149	ok
44 国际部办公楼	54 二校门外停车场	60	220 / 300	141.5	117.9	0.0077 × 10 ⁻¹⁰	5	0.3054	ok
45 第一教学楼	55 大操场	720	1885 / 3000	447.7	373.1	0.00203	4	0.4712	recovered
62 国际部宿舍	40 大操场	400	1885 / 3000	410.5	342.1	3.921 × 10 ⁻¹⁰	4	0.3852	recovered
63 第二教学楼S	41 大操场	400	1885 / 3000	405.5	336.2	0.0077 × 10 ⁻¹⁰	4	0.3806	recovered
67 第一教学楼N	41 大操场	785	2007 / 3000	222.7	149.5	4.904 × 10 ⁻¹⁰	5	0.3274	recovered
68 第二教学楼N	41 大操场	64	2007 / 3000	222.7	149.5	4.904 × 10 ⁻¹⁰	5	0.3274	recovered